

Twist Bioscience

NASDAQ : TWST · Mature-Company DCF
\$10.33B mkt cap · 62M sh · \$172M cash, zero debt · \$377M revenue FY25 (+20%, GM 50.7%); FY26 guide \$442-447M; adj-EBITDA breakeven guided Q4 FY26; Rallied ~+150% over 52wk to ~14x sales while still adj-EBITDA-negative; SBC-heavy (owner-FCF below the headline); ~\$172M net cash; Atlas (DNA data storage) spun out May'25 → external call option

\$165.83

THE CENTRAL QUESTION

Twist tripled over 52 weeks to ~14x sales while still adj-EBITDA-negative — does the platform grow into the price, or has the multiple run past the fundamentals?

Twist Bioscience runs a silicon-based DNA-synthesis platform — miniaturized, high-throughput DNA writing — selling into SynBio and NGS. FY2025 revenue was \$377M (+20%), gross margin 50.7%, adj-EBITDA -\$46.9M and improving; management guides adj-EBITDA breakeven for Q4 FY2026 as the new Wilsonville factory ramps margins. At ~\$87.5 (~\$5.4B cap, net cash) the stock has run ~150% over a year to ~14x sales while still losing money, and SBC keeps owner-FCF below the adj-EBITDA headline. The DCF asks whether 16-24% growth and the margin ramp grow into that price. Weighted fair value \$43.95, ~50% below spot — only the bull and ultra-bull clear the quote.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$43.95

expected fair value today
-73.5% vs spot \$165.83

FORWARD COMPOUNDED VALUE

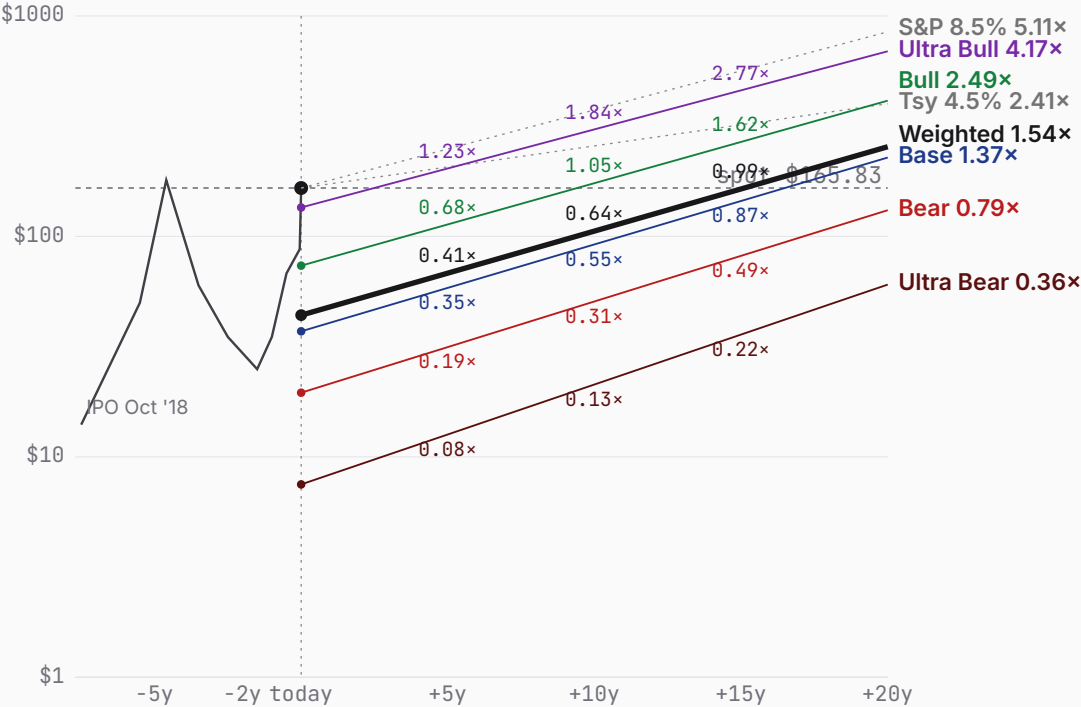
+5y	+10y	+15y	+20y
\$68.15	\$105.75	\$164.22	\$255.20
0.41x spot	0.64x spot	0.99x spot	1.54x spot

WEIGHTING RATIONALE

- Ultra Bear 13% Stall at ~7.6% growth; never durably profitable.
- Bear 27% Decelerate to ~12%; margins up but multiple ran ahead.
- Base 33% ~16% growth + factory margin ramp; modal, ~58% below spot.
- Bull 18% Hold ~20% growth; cost-leader operating leverage.
- Ultra Bull 9% Dominant utility + Atlas; the only case above spot.

Bull (18%) + Ultra Bull (9%) together contribute \$25.44 — 58% of the \$43.95 expected value despite 27% combined probability. Today's spot (\$165.83) sits 277% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	BEAR	BASE	BULL	ULTRA BULL
\$7.50	\$19.55	\$37.12	\$73.69	\$135.31
-95.5% vs spot	-88.2% vs spot	-77.6% vs spot	-55.6% vs spot	-18.4% vs spot
CAGR +7.6% WACC 11.0% CAGR +7.6% Prob 13%	CAGR +12.2% WACC 10.0% CAGR +12.2% Prob 27%	CAGR +16.2% WACC 9.5% CAGR +16.2% Prob 33%	CAGR +20.1% WACC 9.0% CAGR +20.1% Prob 18%	CAGR +24.1% WACC 8.5% CAGR +24.1% Prob 9%
Growth stalls; never profitable	Decelerates; multiple ran ahead	Solid growth; factory margin ramp	High growth; cost leadership	Dominant utility; Atlas windfall

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2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

Twist Bioscience scenario narratives

PROBABILITY WEIGHTS

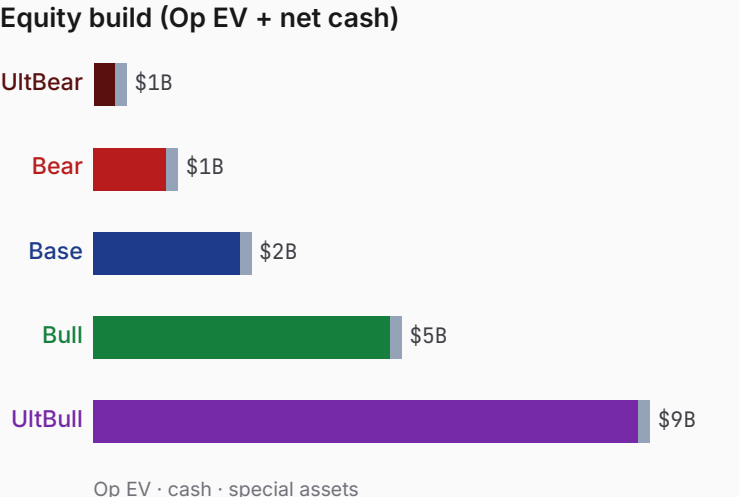
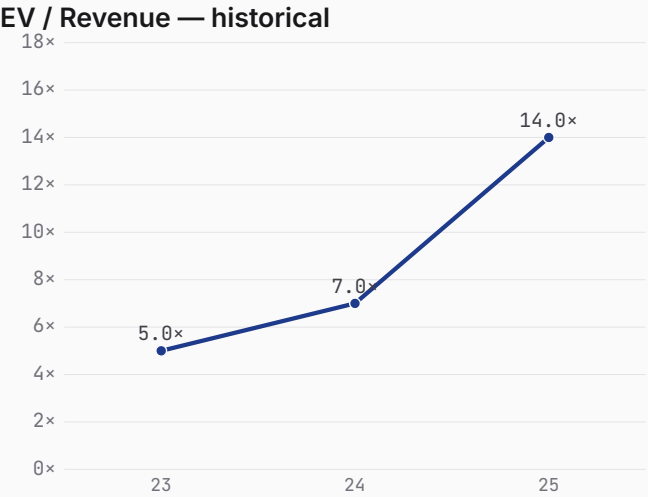
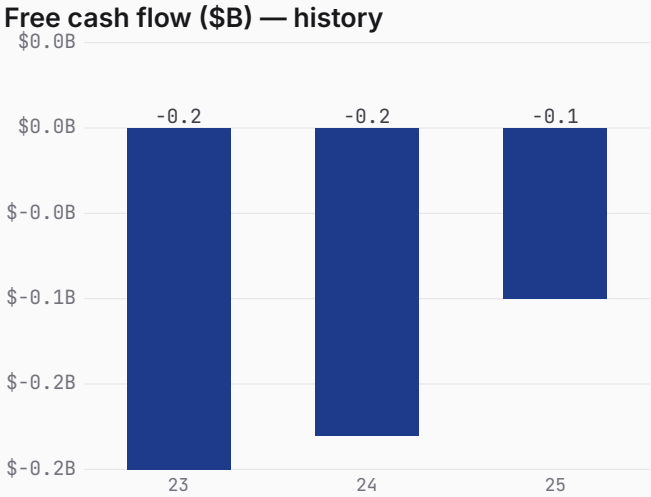
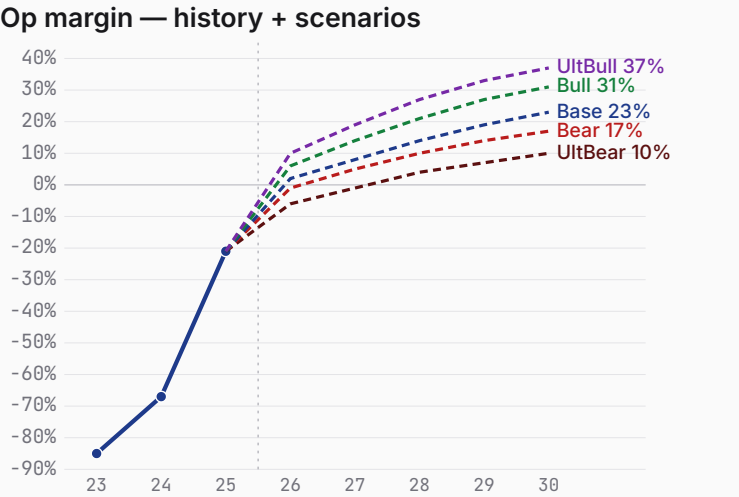
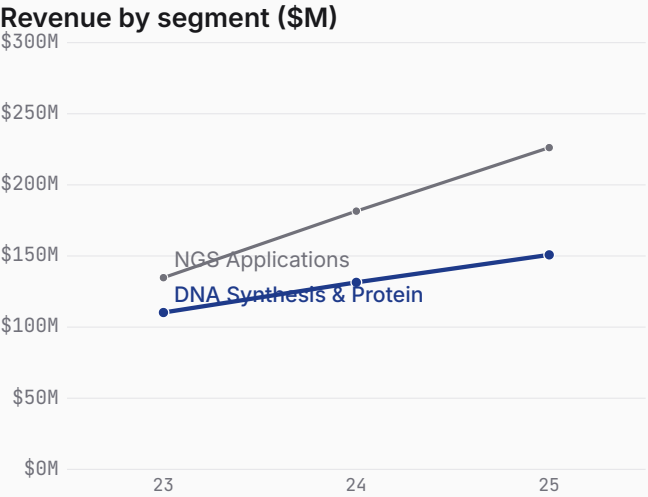
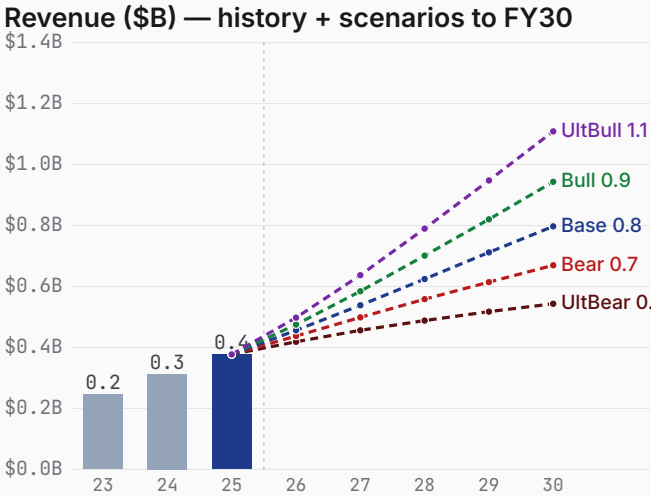
Ultra Bear 13% Bear 27% Base 33% Bull 18% Ultra Bull 9% · Weighted expected \$43.95 (-73.5% vs spot)

ULTRA BEAR	\$7.50	BEAR	\$19.55	BASE	\$37.12	BULL	\$73.69	ULTRA BULL	\$135.31
	-95.5% vs spot		-88.2% vs spot		-77.6% vs spot		-55.6% vs spot		-18.4% vs spot
Probability 13%		Probability 27%		Probability 33%		Probability 18%		Probability 9%	
Growth stalls; never profitable		Decelerates; multiple ran ahead		Solid growth; factory margin ramp		High growth; cost leadership		Dominant utility; Atlas windfall	
WHY 13%		WHY 27%		WHY 33%		WHY 18%		WHY 9%	
13% — assumes commoditization plus a funding winter that stalls the synthesis franchise. Possible given well-capitalized rivals, but cuts against 13 straight sequential-growth quarters.		27% — the natural deceleration path as a 20% grower matures into a low-teens grower against capable competition. The single most defensible outcome alongside the base.		33% — the modal path: 13 sequential growth quarters, the guided breakeven, and a factory built to lift margins. Carries the most weight precisely because it is the management plan working.		18% — requires growth to hold at 20%+ AND the factory to convert into 29% owner-FCF margins. A demanding but coherent cost-leader outcome; the steelman the price needs.		9% — stacks utility-grade dominance, 24% growth, and an Atlas commercialization that is currently an external option Twist only holds equity in. Low-probability by construction; the right tail, not the expectation.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
Revenue compounds only ~7.6% as price competition from deep-pocketed IDT and GenScript caps the synthesis franchise and biotech-funding softness slows orders. The factory's margin ramp arrives but is competed away, so owner-FCF starts negative and barely crosses into positive late in the window.		Growth slows to ~12.2% as the easy share gains mature and competitive pricing pressures the ramp. The Wilsonville factory still drives real gross-margin gains and the company reaches breakeven, but the second-derivative disappoints relative to what ~14x sales embedded.		Revenue compounds ~16.2% — consistent with the raised FY26 guide of \$442-447M extended forward — as Twist holds share across SynBio and NGS and the Factory of the Future delivers its designed margin step. Adj-EBITDA breakeven in Q4 FY2026 converts into sustained, growing operating leverage.		Growth holds ~20.1% as silicon synthesis cements its picks-and-shovels cost advantage and Twist takes share at the expense of higher-cost methods. NGS Applications scales and Protein Solutions adds a higher-value mix, pulling gross margin and operating leverage up together past breakeven into clear profitability.		Twist becomes the default DNA-synthesis utility — the silicon platform's cost and throughput advantage compounds into share dominance across research and applied markets, with platform-grade margins. Growth compounds ~24.1% and the Atlas Data Storage equity stake, spun out in May 2025, commercializes into a real call-option windfall.	
Owner-FCF margin grinds from about -8% to 8% — real but thin. At a ~7.6% grower that never builds durable platform economics, the DCF supports \$7.50, roughly 91% below spot. The 14x-sales multiple was pricing an outcome the business does not reach.		Owner-FCF margin climbs from about -3% to 15%, generating genuine cash flow over time. But a low-teens grower at that terminal margin is worth \$19.55 on the DCF, roughly 78% below spot — the business executes, the entry multiple does not survive contact with decelerating growth.		Owner-FCF margin ramps from roughly 0% to 21%, so the platform finally throws off real cash. Yet the modal value is \$37.12, about 58% below spot. This is the crux — even the company executing its plan well leaves a wide gap, because the price already capitalized the success and then some.		Owner-FCF margin reaches about 29% as fixed factory costs lever across a much larger revenue base. A durable 20% grower with platform margins supports \$73.69, still ~16% below spot — the bull case nearly justifies the quote but does not fully clear it.		Owner-FCF margin reaches ~35% as a dominant utility with pricing power. This combination supports \$135.31, about 55% above spot — the only scenario that materially beats the price, and it requires both the utility outcome and the data-storage option paying off.	

Twist Bioscience business snapshot

Bear \$19.55 Base \$37.12 Bull \$73.69

FY23-FY25 history + FY26-FY30 scenario projections · fiscal years end Sept 30 · FY2025 + Q2 FY2026 results



Sources: Twist Bioscience FY2025 (Sept 30) + Q2 FY2026 results (CIK 1581280), FY26 guidance. Revenue as a growth-rate path off FY25; FCF = revenue x an SBC-adjusted owner-FCF margin (starts negative — SBC keeps owner-FCF below the guided adj-EBITDA breakeven); Gordon terminal at scenario WACC. EV/sales vs DNA-synthesis / life-science-tools peers.

Twist Bioscience 7 Powers · the moat, scored

Bear \$19.55 Base \$37.12 Bull \$73.69

Arena. High-throughput DNA synthesis (plus NGS tools) — Twist's silicon platform versus IDT/Danaher, GenScript, and enzymatic-synthesis challengers Ansa and DNA Script.

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies 	Silicon chip + the Wilsonville factory spread fixed cost across high volume — the core cost advantage, still ramping.
Network Economies 	Essentially none — a DNA supplier's value does not rise with other customers' usage.
Counter-Positioning 	Silicon synthesis is a structurally different, lower-cost method legacy column players cannot easily match without cannibalizing.
Switching Costs 	Some workflow and validation lock-in for NGS and protein customers, but DNA is largely a fungible input across vendors.
Branding 	Strong quality reputation for synthesis fidelity, but buyers are price- and spec-driven, not brand-driven.
Cornered Resource · thesis leans here 	The proprietary silicon-synthesis process and IP — the throughput-per-cost engine rivals have not replicated at scale.
Process Power 	Accumulated manufacturing know-how in miniaturized synthesis, but a process advantage rather than an unassailable one.

PEER SET

IDT (Danaher) Deep-pocketed synthesis leader inside Danaher; can fund price competition	· division of Danaher
GenScript Broad gene-synthesis and CRO scale; mid-teens grower	15% gr · public, well-funded
Ansa Biotechnologies Enzymatic synthesis — a potentially lower-cost long-DNA method	· venture-backed
DNA Script Enzymatic benchtop synthesis; alternative platform path	· venture-backed

THREAT VECTORS · FALSIFIERS

scale_economies / pricing · IDT (Danaher), GenScript
falsifier: Gross margin failing to advance toward platform levels as well-capitalized rivals compete on price during the factory ramp.

cornered_resource erosion · Ansa, DNA Script
falsifier: Enzymatic synthesis reaching commercial cost/throughput parity, neutralizing the silicon platform's structural advantage.

optionality never lands · Atlas Data Storage
falsifier: DNA data storage failing to commercialize, leaving the spun-out equity stake worth little and removing the right-tail.

COMPETITIVE READ

Twist holds a real but medium-durability Power: a cornered silicon-synthesis process backed by scale economies the Wilsonville factory is still building. The audit verdict is a credible cost-leader, not yet an unassailable moat — DNA is partly fungible and a Danaher-owned IDT can fund a price war. The falsifier the bull must clear is the gross-margin ramp: if margins advance toward platform levels while 20%+ growth holds, the Power is durable; if pricing competition flattens the margin curve, it is a strong franchise priced as if it were a moat.

Twist Bioscience show your work

Bear \$19.55 Base \$37.12 Bull \$73.69 · Weighted \$43.95 (-73.5%)

ULTRA BEAR					\$7.50	BEAR					\$19.55	BASE					\$37.12	BULL					\$73.69	ULTRA BULL					\$135.31						
ASSUMPTIONS						ASSUMPTIONS						ASSUMPTIONS						ASSUMPTIONS						ASSUMPTIONS											
5y revenue CAGR (%)					+7.6	5y revenue CAGR (%)					+12.2	5y revenue CAGR (%)					+16.2	5y revenue CAGR (%)					+20.1	5y revenue CAGR (%)					+24.1						
Year-5 op margin (%)					10.0	Year-5 op margin (%)					17.0	Year-5 op margin (%)					23.0	Year-5 op margin (%)					31.0	Year-5 op margin (%)					37.0						
WACC (%)					11.0	WACC (%)					10.0	WACC (%)					9.5	WACC (%)					9.0	WACC (%)					8.5						
Terminal growth (%)					3.0	Terminal growth (%)					3.5	Terminal growth (%)					4.0	Terminal growth (%)					4.5	Terminal growth (%)					5.0						
Terminal FCF (\$B)					\$0.04B	Terminal FCF (\$B)					\$0.10B	Terminal FCF (\$B)					\$0.17B	Terminal FCF (\$B)					\$0.27B	Terminal FCF (\$B)					\$0.39B						
Starting revenue (\$B)					0.38	Starting revenue (\$B)					0.38	Starting revenue (\$B)					0.38	Starting revenue (\$B)					0.38	Starting revenue (\$B)					0.38						
Scenario probability (%)					13	Scenario probability (%)					27	Scenario probability (%)					33	Scenario probability (%)					18	Scenario probability (%)					9						
5-YEAR DCF · \$B						5-YEAR DCF · \$B						5-YEAR DCF · \$B						5-YEAR DCF · \$B						5-YEAR DCF · \$B											
FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF		FY	Rev	Margin	FCF	PV FCF	
FY26	0.42	-6%	-0.03	-0.03		FY26	0.44	-1%	-0.01	-0.01		FY26	0.46	+2%	+0.00	+0.00		FY26	0.48	+6%	+0.02	+0.02		FY26	0.50	+10%	+0.04	+0.04		FY26	0.50	+10%	+0.04	+0.04	
FY27	0.46	-1%	-0.01	-0.01		FY27	0.50	+5%	+0.01	+0.01		FY27	0.54	+8%	+0.03	+0.03		FY27	0.58	+14%	+0.07	+0.06		FY27	0.64	+19%	+0.11	+0.09		FY27	0.64	+19%	+0.11	+0.09	
FY28	0.49	+4%	+0.01	+0.01		FY28	0.56	+10%	+0.04	+0.03		FY28	0.62	+14%	+0.07	+0.06		FY28	0.70	+21%	+0.13	+0.10		FY28	0.79	+27%	+0.20	+0.15		FY28	0.79	+27%	+0.20	+0.15	
FY29	0.52	+7%	+0.03	+0.02		FY29	0.61	+14%	+0.07	+0.05		FY29	0.71	+19%	+0.12	+0.08		FY29	0.82	+27%	+0.20	+0.14		FY29	0.95	+33%	+0.29	+0.21		FY29	0.95	+33%	+0.29	+0.21	
FY30	0.54	+10%	+0.04	+0.03		FY30	0.67	+17%	+0.10	+0.06		FY30	0.80	+23%	+0.17	+0.11		FY30	0.94	+31%	+0.27	+0.18		FY30	1.11	+37%	+0.39	+0.26		FY30	1.11	+37%	+0.39	+0.26	
Σ PV explicit FCF					+0.01	Σ PV explicit FCF					+0.15	Σ PV explicit FCF					+0.27	Σ PV explicit FCF					+0.50	Σ PV explicit FCF					+0.75						
+ PV terminal (g 3.0%)					0.33	+ PV terminal (g 3.5%)					0.99	+ PV terminal (g 4.0%)					2.01	+ PV terminal (g 4.5%)					4.12	+ PV terminal (g 5.0%)					7.74						
EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE						EQUITY BUILD · \$B & PER SHARE											
Operating EV					\$0.34B	Operating EV					\$1.14B	Operating EV					\$2.28B	Operating EV					\$4.62B	Operating EV					\$8.49B						
+ Cash & investments					\$0.17B	+ Cash & investments					\$0.17B	+ Cash & investments					\$0.17B	+ Cash & investments					\$0.17B	+ Cash & investments					\$0.17B						
= Equity value					\$0.51B	= Equity value					\$1.31B	= Equity value					\$2.45B	= Equity value					\$4.79B	= Equity value					\$8.66B						
FY30 shares (M)					68	FY30 shares (M)					67	FY30 shares (M)					66	FY30 shares (M)					65	FY30 shares (M)					64						
= Expected per share					\$7.50	= Expected per share					\$19.55	= Expected per share					\$37.12	= Expected per share					\$73.69	= Expected per share					\$135.31						
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT						FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT											
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		
\$12.64	\$21.30	\$35.88	\$60.47		\$31.49	\$50.71	\$81.67	\$131.52		\$58.44	\$91.99	\$144.82	\$227.98		\$113.38	\$174.45	\$268.41	\$412.99		\$203.46	\$305.93	\$460.02	\$691.71		\$12.64	\$21.30	\$35.88	\$60.47		\$31.49	\$50.71	\$81.67	\$131.52		
0.08×	0.13×	0.22×	0.36×		0.19×	0.31×	0.49×	0.79×		0.35×	0.55×	0.87×	1.37×		0.68×	1.05×	1.62×	2.49×		1.23×	1.84×	2.77×	4.17×		0.08×	0.13×	0.22×	0.36×		0.19×	0.31×	0.49×	0.79×		

Twist Bioscience

People · Opportunity · Context · Deal

Bear \$19.55 Base \$37.12 Bull \$73.69

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Emily Leproust · founder-led

13 yrs in seat

2.4%

insider ownership

MEDIUM

key-person risk

Capital allocation (realized)
Co-founder Emily Leproust has been CEO since 2013 and Chair since 2018; co-founder Bill Banyai remains an executive and director (the DNA-data-storage "Atlas" spinout). The synthetic-DNA business is past early stage (~\$377M revenue, +20%) and targeting adjusted-EBITDA breakeven by late FY2026, on ~\$232M cash. Funding is via an active \$200M ATM (ongoing dilution); no dividend or buyback.

Incentive alignment
Leproust's FY2025 compensation was ~\$13.8M (equity-heavy; up on a large stock grant), with ~94% say-on-pay support. Insiders hold only ~2.4% of the company; alignment is through equity awards, not a founding control block. Insider trading is net selling under a 10b5-1 plan (much of it RSU tax withholding), no open-market buying.

GOVERNANCE FLAGS

- single class of common stock, one vote per share — no dual-class or super-voting; no controlling shareholder (largest holder ARK ~12%)
- combined Chair & CEO (Leproust), mitigated by a lead independent director (Robert Chess)
- classified / staggered board (three classes) — an entrenchment feature
- 7 of 9 directors independent; ongoing ATM dilution with no buyback to offset
- a securities class action (following a short-seller report) was allowed to proceed — an open litigation flag
- director interlock: a sitting director (Ragusa) is CEO of another company in the watchlist (GRAIL)

PEOPLE READ

A clean one-share-one-vote register with no controlling holder, a 7-of-9 independent board, and a lead independent director are the positives. Holding it at 3 are a combined Chair/CEO, a classified board, ongoing ATM dilution, and an open securities class action. Key-person risk is moderate — a founder Chair/CEO, but a second co-founder still active, an institutionalized board, and a business past early stage (~\$377M revenue).

THE FOUR LEGS

People

observable composite · 1–5

Opportunity

The scenario distribution · the business snapshot · the Arthur Indicator

Context

The 7 Powers analysis (Power Audit)

Deal

Open-market common — entry price weighed against the probability-weighted fair value (the -49.8% finding) + position sizing.

Twist Bioscience

supporting analysis · disclaimers · glossary

Bear \$19.55 Base \$37.12 Bull \$73.69

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

Cost-leader picks-and-shovels
A cheaper way to write DNA — a 51%-GM platform selling shovels to every genomics buyer, not one drug.
- 2

Real breakeven inflection
Q4 FY2026 adj-EBITDA breakeven is real; 13 sequential growth quarters and the Wilsonville factory make the margin ramp credible.
- 3

Free Atlas optionality
Data storage spun out as Atlas in May 2025; Twist holds equity, so any commercialization is upside the DCF ignores.
- 4

Multiple, not the business
The bear case is about price, not execution — at ~14x sales even a well-run 16% grower is worth far less than spot.
- 5

SBC gap to owner-FCF
Adj-EBITDA flatters: heavy stock comp keeps owner-FCF below the headline, so 'breakeven' precedes positive owner cash flow.

FALSIFICATION TRIGGERS

Bull validation Growth sustained at 20%+ with adj-EBITDA durably positive; gross margin marching toward platform levels; Atlas gaining traction.	Bear validation Growth decelerating toward low-teens; the margin ramp stalling under IDT/GenScript pricing; SBC keeping owner-FCF negative well past the guided breakeven.	Reframe needed Sustained price competition on gross margin, or a biotech-funding winter cutting orders, moves this from rich-but-executing to challenged.
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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Silicon DNA synthesis — Twist's core platform — DNA written on a miniaturized silicon chip, enabling high throughput at low cost per base versus older column methods.	SynBio / NGS segments — The two demand areas: synthetic biology (DNA Synthesis and Protein Solutions) and next-generation sequencing applications (target enrichment, library prep).	Factory of the Future — The new Wilsonville, Oregon plant built to scale capacity and step up gross margin — the engine behind the modeled margin ramp.
Atlas (DNA data storage) — The DNA-data-storage venture spun out as independent Atlas Data Storage in May 2025; Twist retains equity, so it is now an external call option, not in-house.	Adj-EBITDA vs owner-FCF — Adj-EBITDA adds back stock comp (breakeven Q4 FY2026); owner-FCF subtracts real SBC + capex, so it turns positive later.	SBC — Stock-based compensation — a real cost to owners via dilution that adj-EBITDA excludes; deducted here from owner FCF.